ICSI 418Y

Dr. Shashank Arora

Team 5 – Milestone 2: Requirements Analysis

1. Project Overview and Users

Project Title:

- ParkSmart: Campus Parking Availability and Management System

Team Members:

- Leigha Schilling
- Kanza Fatima
- Denise Ortiz
- Nelson Hudgins
- Yuhan Sun

Project Overview:

ParkSmart is a web-based application designed to help commuter students at UAlbany find available parking spots more easily. Students can view commuter parking lots and their reported availability, select an available spot, and reserve it when they arrive. When they leave, they can unreserve the spot so it becomes available to another student. The goal of ParkSmart is to reduce the time commuter students spend searching for parking.

Changes from Milestone 1:

There are currently no changes to the MVP features or optional stretch goals from Milestone 1.

User Roles/Personas:

Commuter Student:

The main users of ParkSmart are UAlbany commuter students with commuter parking permits. Their goal is to quickly find and reserve an available commuter parking spot and unreserve it when they leave.

Parking Services:

The university's Parking Services department is the customer for ParkSmart. Their goal is to provide accurate parking lot and parking spot information for the system.

Unresolved Assumptions and Questions:

The system assumes that students will unreserve their parking spot when they leave. If they forget, the availability shown in the system may be inaccurate.

Another question is how the system will verify that a student has arrived at the selected parking lot before allowing them to reserve a spot. Location tracking could address this, but it is currently an optional stretch goal.

2. User Stories and Acceptance Criteria

MVPS:

1. License plate entry
2. View commuter parking lots
3. View parking-lot capacity
4. View initial parking spots

5. Reserve a parking spot
6. Unreserve a parking spot

User Stories:

The following user stories cover all features included in the ParkSmart minimum viable product.

US-01 Enter and Validate a License Plate

Label: MVP

Feature: License plate entry

User Story: As a commuter student, I want to enter my license plate number so that my parking reservation can be connected to my vehicle.

Acceptance Criteria:

1. When the student submits a nonblank license plate in the application's accepted format, the system saves it for use with the student's reservation.

2. If the license plate field is blank or has an invalid format, the system rejects the submission and explains how to correct it.

3. The student can review the accepted license plate before continuing to the parking-lot list.

US-02 View Commuter Parking Lots

Label: MVP

Feature: View commuter parking lots

User Story: As a commuter student, I want to view the available commuter parking lots so that I can choose where to look for a parking space.

Acceptance Criteria:

1. The system displays every commuter parking lot currently available in ParkSmart.

2. Each displayed lot includes its name and current reported availability.

3. If no parking-lot info is available, the system displays an explanatory message instead of an empty misleading list.

US-03 View Parking Lot Capacity

Label: MVP

Feature: View parking lot capacity

User Story: As a commuter student, I want to see the capacity and reported availability of each parking lot so that I can compare lots before selecting one parking lot.

Acceptance Criteria:

1. Each displayed lot shows its total capacity and the number of currently available spaces.

2. A successful reservation decreases the selected lot's available-space count by one, and a successful release increases it by one. Rejected or repeated requests do not incorrectly change the count.

3. A lot with zero available spaces is clearly shown as full. If availability cannot be retrieved, the system displays it as unknown rather than showing an incorrect count.

US-04 View Individual Parking Spots

Label: MVP

Feature: View Individual Parking Spots

User Story: As a commuter student, I want to view individual parking spaces in a selected lot so that I can identify a space available for reservation.

Acceptance Criteria:

1. After the student selects a lot, the system displays the identifiers and application-recorded statuses of the spaces belonging to that lot.

2. Available and reserved spaces are distinguishable through text labels, and only spaces shown as available can be selected for a new reservation.

3. If no spaces are available, the system displays a clear message and allows the student to return to the parking-lot list.

US-05 Reserve an Available Parking Spot

Label: MVP

Feature: Reserve a parking spot

User Story: As a commuter student, I want to reserve an available parking space after arriving at the lot so that my selected space is recorded as reserved for my vehicle.

Acceptance Criteria:

1. When the student confirms an available space with an accepted license plate, the system saves the reservation, marks the space as reserved, updates the lot's availability, and displays a reservation confirmation.

2. If the space is no longer available when the request is processed, the system rejects the reservation, explains the conflict, and allows the student to choose another space. Simultaneous requests for the same space cannot both succeed.

3. Repeating the same reservation request does not create duplicate reservations or decrease the available-space count more than once.

US-06 View Reservation Confirmation

Label: MVP

Feature: Reserve a parking spot — review reservation details

User Story: As a commuter student, I want to view the confirmation and details of my current reservation so that I can verify which parking space is associated with my vehicle.

Acceptance Criteria:

1. The reservation confirmation displays the saved parking-lot name, space identifier, license plate, and reservation status.

2. When the student refreshes or returns to the reservation view during the same session, the system retrieves the saved reservation details without creating another reservation or changing availability.

3. If the student has no active reservation, the system displays a clear message. If reservation details cannot be retrieved, the system displays an error and allows the student to retry.

US-07 Unreserve a Parking Spot

Label: MVP

Feature: Unreserve a Parking Spot

User Story: As a commuter student, I want to release my reserved parking space when I leave so that the application shows it as available to another student.

Acceptance Criteria:

1. The student can request to release a space from their current reservation. The system asks for confirmation and identifies the lot and space being released. Canceling leaves the reservation unchanged.

2. Confirming the release ends the reservation, marks the space as available, updates the lot's available-space count, and displays a release confirmation.

3. A release request for another student's reservation is rejected. Releasing an already released reservation does not increase the available-space count again.

3. Five Critical Use Cases

UC-01 Enter License Plate

Related User Story: US-01

Why This Use Case Is Critical:

This use case is critical because the student's license plate connects their vehicle to a parking reservation.

Actors:

Primary Actor: Commuter Student

Supporting Actor: ParkSmart System

Trigger:

The student enters their license plate number to begin using ParkSmart.

Preconditions:

- The student has access to ParkSmart.
- The student has a commuter parking permit.

Main Success Flow:

1. The student enters their license plate number.
2. The student submits the license plate number.
3. The system checks that the license plate is not blank and is in the accepted format.

4. The system accepts and saves the license plate.

5. The system allows the student to continue to the commuter parking lot list.

Alternative and Error Flows:

- **Step 3:** If the license plate is blank or has an invalid format, the system rejects it and explains how to correct it. The student can correct the license plate and try again.

Postconditions:

Success: The license plate is saved and can be connected to the student's parking reservation.

Failure: No license plate is saved, and the student remains on the license plate entry step.

UC-02 View and Select a Commuter Parking Lot

Related User Stories: US-02, US-03

Why This Use Case Is Critical:

This use case is critical because it allows students to compare commuter parking lots and choose one based on its reported availability.

Actors:

Primary Actor: Commuter Student

Supporting Actor: ParkSmart System

Trigger:

The student continues to the commuter parking lot list after entering a valid license plate.

Preconditions:

The student's license plate has been accepted.

Commuter parking lot information is available in the system.

Main Success Flow:

1. The system displays the available commuter parking lots.
2. The system shows the name and reported availability of each lot.
3. The system shows the capacity and number of available spaces for each lot.
4. The student compares the commuter parking lots.
5. The student selects a parking lot.
6. The system displays the individual parking spaces in the selected lot.

Alternative and Error Flows:

- **Step 1:** If parking lot information is unavailable, the system displays an explanatory message, and the workflow ends.
- **Step 3:** If availability cannot be retrieved for a lot, the system shows its availability as unknown instead of displaying an incorrect count. The student can continue viewing the other lots.

Postconditions:

Success: The student has selected a commuter parking lot and can view its parking spaces.

Failure: No parking lot is selected, and no reservation is created.

UC-03 View Available Parking Spots

Related User Story: US-04

Why This Use Case Is Critical:

This use case is critical because students need to see which parking spaces are available before choosing one to reserve.

Actors:

Primary Actor: Commuter Student

Supporting Actor: ParkSmart System

Trigger:

The student selects a commuter parking lot.

Preconditions:

- The student has entered a valid license plate.
- The student has selected a commuter parking lot.

Main Success Flow:

1. The student selects a commuter parking lot.
2. The system retrieves the parking spaces for the selected lot.
3. The system displays the individual parking spaces.
4. The system shows which spaces are available and which are already reserved.
5. The student reviews the available parking spaces.
6. The student selects an available parking space.

Alternative and Error Flows:

- **Step 2:** If the parking spot information cannot be retrieved, the system displays an explanatory message, and the workflow ends.

- **Step 4:** If there are no available parking spaces, the system shows that the lot is full. The student can return to the parking lot list and choose another lot.

Postconditions:

Success: The student has selected an available parking space.

Failure: No parking space is selected, and no reservation is created.

UC-04 Reserve a Parking Spot

Related User Stories: US-05, US-06

Why This Use Case Is Critical:

This use case is critical because reserving an available parking spot is one of the main purposes of ParkSmart.

Actors:

Primary Actor: Commuter Student

Supporting Actor: ParkSmart System

Trigger:

The student selects an available parking spot and chooses to reserve it after arriving at the parking lot.

Preconditions:

- The student has entered a valid license plate.
- The student has selected a commuter parking lot.
- The selected parking spot is available.
- The student has arrived at the parking lot.

Main Success Flow:

1. The student selects an available parking spot.
2. The student chooses to reserve the parking spot.
3. The system checks that the parking spot is still available.
4. The system creates the reservation and connects it to the student's license plate.
5. The system marks the parking spot as reserved.
6. The system updates the parking lot's reported availability.
7. The system displays a confirmation showing the parking lot, parking spot, and license plate.

Alternative and Error Flows:

- **Step 3:** If the parking spot is no longer available, the system rejects the reservation and informs the student. The student can select another available spot.
- **Step 4:** If the reservation cannot be completed, the system does not mark the spot as reserved and informs the student that the reservation was unsuccessful.

Postconditions:

Success: The parking spot is reserved for the student's vehicle, and the student can view the reservation confirmation.

Failure: No reservation is created, and the parking spot remains in its previous state.

UC-05 Unreserve a Parking Spot

Related User Story: US-07

Why This Use Case Is Critical:

This use case is critical because unreserving a parking spot when a student leaves makes the spot available to other commuter students.

Actors:

Primary Actor: Commuter Student

Supporting Actor: ParkSmart System

Trigger:

The student is leaving their parking spot and chooses to unreserve it.

Preconditions:

- The student has an active parking reservation.
- The parking spot is connected to the student's license plate.

Main Success Flow:

1. The student views their current parking reservation.
2. The student chooses to unreserve the parking spot.
3. The system asks the student to confirm that they want to unreserve the spot.
4. The student confirms the request.
5. The system removes the reservation.
6. The system marks the parking spot as available.
7. The system updates the parking lot's reported availability.
8. The system confirms that the parking spot has been unreserved.

Alternative and Error Flows:

- **Step 4:** If the student cancels the request, the reservation remains active and the workflow ends.

- **Step 5:** If the system cannot complete the request, the reservation remains active and the system informs the student that the request was unsuccessful.

Postconditions:

Success: The reservation is removed, and the parking spot becomes available to another commuter student.

Failure: The reservation remains active, and the parking spot remains reserved.

4. Nonfunctional Requirements

NFR-01: Parking Availability

Category: Performance

Requirement: ParkSmart should display the available parking spaces within 2 seconds after student selects a parking lot.

Reason: Students should be able to check parking availability quickly, especially when they are trying to get to class on time. Having to wait for information to load can make the process more frustrating.

Verification: Using a test parking lot with 20 spaces and 1 student using the app, check the loading time 5 times. All 5 tests should display the parking information within 2 seconds.

NFR-02: Reservation Accuracy:

Category: Reliability

Requirement: ParkSmart should correctly update the number of open parking spaces whenever a student reserves or unreserves a spot. The system should also prevent 2 students from reserving the same parking spot at the same time.

Reason: Students need accurate parking information, so they don't waste time looking for spaces that are already reserved. If the availability is incorrect, the application wouldn't be helpful.

Verification: Use 2 test users to try reserving the same parking spot at the same time. Only 1 reservation should go through and be successful, and the available space count should decrease by one. Then, unreserve the spot and check that the number of available spaces increases by 1.

NFR-03: Reservation Ownership Protection

Category: Security

Requirement: Only the owning student session shall be allowed to release an active reservation. Knowing another student's license plate or reservation identifier shall not be sufficient to release it. Ownership checks shall apply to every release request.

Reason: Students must not be able to alter another student's reservation or falsely increase parking availability.

Verification: Create a reservation in session A. From separate session B, attempt to release it using its identifier and plate, including a directly submitted request. Each attempt must be rejected with no reservation or count change. Confirm that session A can release it successfully.

NFR-04: Mobile Accessibility

Category: Accessibility

Requirement: ParkSmart should have easy to read text, clear labels, and buttons that are easy to tap on a phone. Parking lots should use green, yellow, and red indicators to show

how full they are. Following this, will be labels such as “Available”, “Limited”, and “Full” so students do not only have to rely on colors alone.

Reason: Students might be in a rush when looking for parking so the application should be easy to understand and navigate quickly, the color indicators will help students quickly compare parking lots, while the labels will make the information easier to understand for students who may have trouble distinguishing colors.

Verification: We can test ParkSmart on 2 different phone screen sizes to check that the text is easy to read, buttons are easy to tap, and each parking lot displays the correct color and label based on how occupied the parking lot is.

NFR-05: Mobile Browser Compatibility

Category: Compatibility

Requirement: ParkSmart should work properly on mobile browsers such as Safari and Chrome. Students should have the ability to view parking availability, reserve their spot of choice, and then unreserve it without display or navigation issues.

Reason: Students may use different types of phones, so we need to make sure ParkSmart works consistently without making the parking process overcomplicated or tedious

Verification: Test ParkSmart using Safari on iPhone and Chrome on an Android phone. On both browsers, we will need to complete the license plate entry, parking lot and spot selection, reservation, confirmation, and unreserve process. We can check that the buttons and text are displayed correctly without any overlapping, cut off features, scrolling issues. This will allow us to record any issues found.

5. **Effort Sharing**

Team size	Joint work	Leigha	Kanza	Denise	Nelson	Yuhan

5	25%	15%	15%	15%	15%	15%

Kanza Fatima:

I worked on Section 1 by completing the Project Overview and Users. I also worked on Section 3 by completing the five critical use cases, UC-01 through UC-05. This included writing the main success flows, alternative and error flows, and postconditions for the main ParkSmart workflows.

Yuhan Sun:

I completed the user stories and acceptance criteria, US-02 through US-07, in Section 2. I also worked with Denise Ortiz on the nonfunctional requirements in Section 4.

Denise Ortiz:

I worked on Section 4 by adding NFR-01 and NFR-02. I worked together with Yuhan Sun and reviewed NFR-04 (Accessibility) and NFR-05 (Compatibility) to better fit ParkSmart's mobile design. This included adding labels for occupancy along the color indicators (Red, Yellow, Green) to make parking availability easier to understand for students with visual disabilities.